

The Utility Problem

Losses are part of every electrical system. They are divided into technical and non-technical. Technical losses occur in any transmission of electricity and are caused by the physical effects of electricity. Technical losses include corona losses, leakage current losses, dielectric losses, Joule losses on lines, cables and transformers, transformer no-load losses and others. Non-technical losses, sometimes also called commercial losses, include, for example, unmeasured consumption, errors in metering, billing or accounting, consumption below the sensitivity limit of meters, unauthorized consumption, etc.

Utilities usually only know the volume of losses for the entire grid and usually estimate the values for smaller parts of the grid. Utilities also do not know the ratio between technical and non-technical losses. However, to detect non-technical losses, it is necessary to know where these losses occur, and therefore, it is not sufficient to know the losses only for the whole grid.

For this purpose, Energy Balancing is a well-known approach. Energy Balancing is a technique of energy measurements comparison between:

- A. Net energy value within the specific grid area - often called segment - measured by grid sensors and;
- B. Net value of the energy consumption on the grid edge - on meter/consumer level.

However, the systematic grid-wide deployment of energy balancing using software tools is not easy. Moreover, if the utility wants to simplify the whole process of implementing energy balances on the electricity grid, it is useful to integrate data from a system containing links & relations between grid elements - GIS data. Without them, it is not possible to automatically create segments and create assignments of consumption points to grid energy measurement devices for accurate energy balancing.

Reducing losses is very important for the company's profits. Losses have to be purchased at a certain market price, and therefore, the company's profits are reduced. If losses are reduced, the company's profits increase and financial resources can be spent on grid modernization.

The Data Solutions and Methods

Awesense's Energy Transition Platform uses a connection of geospatial (GIS) and time series data (meter and grid sensor) to facilitate a systematic approach to the creation and evaluation of the energy balances in the grid. This unique connection helps in automatically creating segments with the assignment of the correct customer locations to the grid measuring energy sensor.

The key datasets for energy balancing are:

- Geospatial Information System (GIS) data:

- For the representation of the relation between the elements, mainly consumption point vs. grid measuring device.
 - For the provision of additional information about consumers and grid elements impacting the energy balance.
- Sensor and its time-series data with information about GIS location:
 - Sensor GIS location for the definition of energy balance segments.
 - Sensor time series for quantification of net energy flows (inflow minus outflow) in the segments.
- Meter time series data:
 - For quantification of net energy consumption (consumption - generation) on grid edge - energy meter/consumer.

As mentioned above, the calculation of the energy balance requires measured data from sensors (e.g. quality meters, portable sensors, other IoT, etc.), which are compared with measured data from grid edge/consumption points. The Awesense data engine can integrate any sensors from any manufacturer. It can also manage different time granularity measurements. If time series measurements at the consumption points are not available, then it is possible to create time series values based on historical consumption and consumption point parameters using machine learning with varying levels of complexity.

There are two basic types of energy balancing, both fully supported by the Awesense Platform:

- The first type uses already integrated sensors in the grid. These are static sensors such as SCADA sensors, quality meters, transformer smart meters, etc. Based on this data, the Awesense Platform is able to calculate the system-level energy balances. Due to the integration of existing sensors in the grid, there is no need to install additional sensors in the grid, and thus, the workload of this solution is greatly reduced while significant results are achieved.
- The second supported type uses portable sensors. With this type, the sensors are deployed in selected areas of the grid. The deployment areas can be identified by line technicians or can be high-loss areas from the example above (this methodology is called the drill-down approach).

After the energy balance calculation, the energy balance can be displayed directly in Awesense True Grid Intelligence (TGI) frontend solution. The energy balance instance, called “Investigation”, contains multiple components:

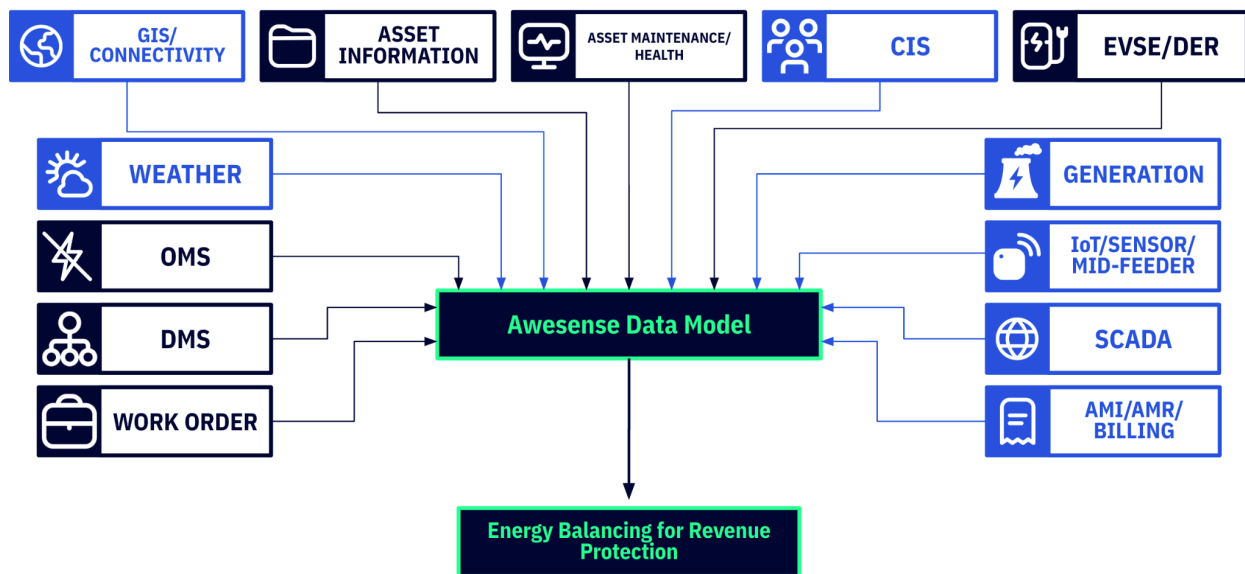
- A map, which shows the calculated segments. Segments are colour-coded according to the results of this energy balance.
- Energy balance graph with different display options. It is possible to calculate the balance for kWh or Ah, display the balance for individual phases or for all phases, choose a bar or line graph, add a graphical display of weekends and nights, add a temperature display, etc. All this helps to add context and makes the energy balance conclusion easier to evaluate.

- Segments can also be displayed in greater detail on a separate page, called “Report View”. It is possible to display statistics of consumption points and measurement sensor data statistics; it is also possible to create custom graphs, statistics and more.

The quantification of the losses is thus done very accurately using the Awesense Platform energy balance. Awesense Platform and its tools provide a systematic and manageable approach to energy balancing. With these tools, utility professionals can run dedicated non-technical loss mitigation programs with ease and stay focused on the goal of reducing energy losses.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Energy Balancing functionality. Consult [instructions](#) about how to access this use case in TGI. If you do not have access to a server where the use case functionality is available, contact [Awesense](#) to see a demo of it.